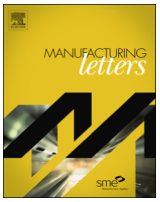
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CRONet: A Convolutional Recurrent Operator Approximator Network to Accelerate Topology Optimization

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Abstract

The convergence of emerging additive and hybrid manufacturing and publicly accessible CAD/CAM datasets has paved the way for social manufacturing, a paradigm where users can access, adapt, and produce designs on demand. This approach often involves leveraging topology optimization—a computational technique that optimally allocates materials within a design space based on performance criteria, such as compliance or thermal resistance. However, traditional methods rely heavily on finite element analysis (FEA) for performance evaluation, which can be computationally intensive, especially for complex high-resolution models, limiting rapid prototyping and iterative design processes essential to advancing social manufacturing and empowering enthusiasts and hobbyists. Recent advancements in operator learning offer an efficient alternative by directly learning mappings between infinite-dimensional spaces, e.g., differential operators, significantly reducing computational time. This research introduces the Convolutional Recurrent Operator Approximator Network (CRONet) as a surrogate model for FEA in topology optimization. CRONet efficiently approximates 2D and 3D operator mappings, enabling rapid, high-fidelity optimization with reduced computational costs. CRONet integrates both convolutional and recurrent neural architectures to capture spatial correlations and temporal dependencies within the optimization process. The recurrent component is crucial as it allows CRONet to learn from the sequential nature of topology optimization iterations, effectively capturing temporal relationships between successive design updates. This enables CRONet to adapt and maintain accurate predictions across iterations, further improving efficiency and robustness. Through benchmark case studies—the Messerschmitt-Bölkow-Blohm (MBB) beam and the cantilever beam—CRONet demonstrates up to a 78% reduction in computational time while retaining compliance accuracy within 5% of traditional FEA solutions. This novel approach promises a transformative impact on the design workflow in AM, making high-fidelity optimization accessible for complex, manufacturable structures.

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Keywords: Topology Optimization; Additive Manufacturing; Deep Operator Network; FEA Surrogate; Accelerated Design

1. Introduction

The widespread adoption of 3D printing together with the increasing availability of CAD/CAM datasets on public repositories has created a new opportunity for social manufacturing—an emerging paradigm where individuals of different skill levels could retrieve existing designs and manufacture them on demand. Rather than starting from scratch, this new paradigm enables manufacturers to use existing designs as the initial version and subsequently iterate them to fit their needs. Often, this adaptation is achieved through topology optimization methods, which involve the optimal distribution of materials within a predefined design space, subject to specific performance criteria and constraints [6] such as compliance using linear elasticity equations [20], thermal properties using heat conduction equation [25], and diffusion or advection-diffusion equations for mass transfer. The underlying governing equations are solved using methods of finite element analysis [36]. However, these computations can be time-consuming, especially when dealing with high-resolution complex models [10] that capture the intri-

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cate details, requiring several hours to a few days on GPUs. For instance, stress analysis of a 14-inch pressure nozzle takes 3.2 days on an NVIDIA GPU [17, 24]. Furthermore, the need for fine discretization to accurately represent complex geometries further exacerbates the computational burden. These challenges make it difficult to screen hundreds of structures in a design family, let alone optimize them [11].

This bottleneck in solving the linear elasticity equations has also hindered the acceleration of structure optimization, both at the micro and macro scale [33, 19]. Over the years, significant developments have occurred in this domain, beginning with the homogenization method [3], the density method [4, 41], phase field method [40], and evolutionary methods [42, 39]. However, these methods suffer from the general problem of mesh dependency leading to a checkerboard pattern. Various filtering and higher-order finite element approaches have been developed to overcome these limitations [14, 26, 44]. Nonetheless, they require significant parameter tuning and often lead to highly complex geometries that are not manufacturable.

To overcome these challenges, adaptive meshing [2, 1], parallel processing [13, 37], and machine learning emerged in recent years. For instance, Deng et al. [11] trained a neural network to predict material behavior by mapping design structures to property. Another approach is to solve costly finite element methods at a coarser mesh and then use surrogate models such as Gaussian processes or neural networks to obtain high-fidelity solutions at finer meshes [9]. However, using machine learning approaches to bypass costly finite element computations requires training on large datasets (e.g., 10^6 geometries in [43] and 10^4 designs in [11]) that are not feasible to obtain without performing costly simulations first and yet still require several hours for prediction (e.g., 3-7 hours for stress field at different mesh refinements in 3D geometries [9]). These challenges limit the applicability of existing computational models for the social manufacturing of dynamically evolving product requirements.

More recently, level set methods have been developed that implicitly represent the optimization problem using the isosurface of a level set function [29]. Due to its implicit nature, intricate geometries are possible via level-set methods while overcoming the issues related to checkerboard patterns and grayscale regions. Since the first level set approach was developed, several improvements were proposed, such as simulating and growing bubbles [12], methods of topological derivatives [32], and fictitious energy surfaces [40]. Methods based on AI techniques, such as deep neural networks, bypass costly finite element calculations by directly learning the optimal structure/geometry. These methods map the input structures to the desired properties. However, the generation of training data remains a bottleneck, requiring up to 100k samples for 2D geometries [38]). Furthermore, the fundamental challenges of repeatedly solving for the desired property fields using finite elements and costly domain updates have limited the existing structure optimization methods to simpler geometries.

In recent years, advancements in deep learning have offered promising alternatives to traditional computational and machine learning methods. More specifically, operator learning frameworks, such as Deep Operator Networks (DeepONet) [21], have emerged as powerful tools for approximating complex mappings between infinite-dimensional function spaces. Unlike traditional neural networks that approximate functions, operator networks learn to approximate operators (e.g., partial differentials), enabling them to map functions to functions directly. Operator networks have demonstrated the ability to predict solutions to partial differential equations (PDEs) with high accuracy and significantly reduced computational time [22]. By learning the underlying operators, such networks can generalize across different inputs, making them a suitable candidate for predicting structural responses in topology optimization without the need for computationally demanding iterative FEA computations.

Several studies have explored the application of operator networks and similar neural operator frameworks in various engineering fields. Lu et al. [23] introduced multifidelity deep neural operators for efficient learning of PDEs, enabling fast inverse design of nanoscale heat transport systems. Their work demonstrated that neural operators could efficiently handle problems with varying fidelity levels, reducing computational costs. He et al. [16] proposed a novel operator network based on the DeepONet architecture to predict stresses in elastoplastic structures with variable complex geometries and loads, demonstrating the network's capability to handle complex material behaviors and loading conditions. Additionally, He et al. [15] developed a point-cloud-based deep operator network for field predictions on 3D parameterized geometries, highlighting the potential for handling complex geometries inherent in AM designs. These advancements indicate the versatility and effectiveness of neural operator frameworks in capturing complex physical phenomena across different domains. Moreover, Ramezankhani et al. [27] presented an advanced physics-informed neural operator for comprehensive design optimization of highly nonlinear systems, illustrating the applicability of these operators in design optimization tasks. Their approach integrated physics-informed constraints directly into the learning process, improving accuracy and generalization in complex engineering systems.

Building upon these recent advancements, this research explores the potential of operator networks to replace traditional finite element analysis (FEA) within the topology optimization process. Specifically, we propose a Convolutional Recurrent Operator Approximator Network (CRONet) as a surrogate model for the computationally intensive FEA simulations. CRONet is designed to learn the complex operator mappings between material distributions, load configurations, and structural displacement fields in both 2D and 3D design domains. Unlike traditional FEA, which requires repetitive computation of governing differential equations, CRONet leverages convolutional and recurrent neural architectures to efficiently capture both spatial and temporal dependencies. CRONet consists of two primary subnetworks: the branch network, which processes material distributions, and the trunk network, which captures load configurations. Additionally, a recurrent component is integrated into the branch network to consider temporal changes during the optimization process. This architecture enables CRONet to replace the FEA solver during most iterations

of the topology optimization loop, providing near-instantaneous predictions of the displacement field while maintaining accuracy within acceptable limits. By leveraging CRONet's ability to approximate complex 2D and 3D operator mappings efficiently, we aim to establish a fast, data-driven approach that retains high accuracy while significantly reducing computational time for predicting structural displacements within the topology optimization process.

The motivation for this work is driven by the demands of additive manufacturing (AM), where intricate geometries and complex material behaviors are common. The proposed methodology seeks to enable rapid design iterations, making high-fidelity optimization practical and accessible even for highly detailed, manufacturable structures. This is particularly advantageous in AM applications, where design flexibility is a key benefit but computational resources often pose a constraint.

The successful integration of CRONet into the topology optimization workflow is demonstrated through two benchmark case studies: the Messerschmitt-Bölkow-Blohm (MBB) beam and the cantilever beam. These studies provide a direct comparison between traditional FEA-based optimization and the CRONet-enhanced approach, highlighting a significant reduction in computational time—up to 78% faster—without compromising the quality of the final optimized designs. The results confirm that CRONet maintains compliance values within 5% of the FEA-derived solutions while achieving faster convergence and improved computational efficiency. This novel approach offers the potential to revolutionize the design workflow in additive manufacturing, enhancing the practicality of high-fidelity optimization for real-world applications.

The remainder of the manuscript is structured as follows: Section 2 outlines the proposed methodology, detailing the formulation of the topology optimization problem and the specifics of the algorithm with DeepONet embeddings. Section 3 presents the case studies used to evaluate the method's performance across distinct design domains and loading conditions. Finally, Section 4 concludes the paper with a summary of key findings and a discussion of future research directions to further enhance this methodology.

2. Methodology

In this section, we present our approach for training and integrating a CRONet model into the topology optimization process. We begin by formulating the classical topology optimization problem, which aims to minimize compliance. Then, we describe the CRONet architecture used as a surrogate for FEA within the optimization framework. Next, we explain the training process for CRONet to accurately predict structural displacement responses across the domain. Finally, we detail how the trained CRONet model is integrated into the optimization loop to streamline the process.

2.1. Problem Formulation

Given a design domain (Ω) discretized using finite elements mesh, the primary objective of the topology optimization is to minimize the structural compliance $c(x)$ (where x represents the design variable) by finding the most efficient material distribution within the domain under prescribed loading and boundary conditions. The optimization problem, based on the Solid Isotropic Material with Penalization (SIMP) approach [3, 4], is formulated as follows following similar notation as [30]:

$$\min_{x} \; c(x) = \mathbf{U}^{\mathrm{T}}\mathbf{K}\mathbf{U} = \sum_{e=1}^{N} (x_e)^p \, \mathbf{u}_e^{\mathrm{T}} \, \mathbf{k}_e \, \mathbf{u}_e, \tag{1}$$

$$\text{subject to:} \quad \frac{V(x)}{V_0} = f,$$

$$\mathbf{K}\mathbf{U} = \mathbf{F},$$

$$\mathbf{0} < \mathbf{x}_{\min} \leq \mathbf{x} \leq \mathbf{1}$$

Here, $\mathbf{U}$ and $\mathbf{F}$ represent the global displacement and force vectors, respectively, while $\mathbf{K}$ denotes the global stiffness matrix. The vector $\mathbf{x}$ contains the design variables (relative density of each element), and $\mathbf{x}_{\min}$ is a vector specifying minimum relative densities chosen to prevent singularities. The terms $\mathbf{u}_e$ and $\mathbf{k}_e$ correspond to the displacement vector and stiffness matrix at the element level. The parameter N represents the total number of elements used to discretize the design space. The parameter p is the penalization factor, commonly set to 3 [30]. The terms $V(x)$ and V_0 refer to the material volume and the volume of the entire design domain, respectively, while f is the specified volume fraction constraint.

The optimization problem in Eq. (1) can be solved using various techniques, Sequential Linear Programming (SLP) methods, Method of Moving Asymptotes (MMA) developed by Svanberg (1987) [35], and the Optimality Criteria (OC) methods. In this work, we opt for the OC method because of its simplicity and efficiency in handling large-scale topology optimization problems. As outlined by Bendsøe (1995) [5], a heuristic update rule as shown in the equation below can be employed to iteratively adjust the design variables, guiding the solution towards optimality.

$$x_e^{\text{new}} = \begin{cases} \max\left(x_{\min},\, x_e - m\right), \\ \quad \text{if } x_e B_e^{\eta} \leq \max\left(x_{\min},\, x_e - m\right), \\ x_e B_e^{\eta}, \\ \quad \text{if } \max\left(x_{\min},\, x_e - m\right) < x_e B_e^{\eta} < \min\left(1,\, x_e + m\right), \\ \min\left(1,\, x_e + m\right), \\ \quad \text{if } \min\left(1,\, x_e + m\right) \leq x_e B_e^{\eta}, \end{cases}$$

where m, known as the move-limit, is a positive value that defines the maximum allowable change for each iteration of topology optimization, and η (commonly set to $1/2$) is a damping coefficient used to stabilize the update process. The term B_e is calculated based on the optimality condition as:

$$B_e = \frac{-\frac{\partial c}{\partial x_e}}{\lambda \frac{\partial V}{\partial x_e}}, \tag{2}$$

where λ is a Lagrange multiplier determined through a bi-sectioning algorithm [30]. The SIMP method is employed to

interpolate the material properties, where the Young's modulus E at point x is given by:

$$E(\mathbf{x}) = E_0\,(\mathbf{x})^p\,, \tag{3}$$

with E_0 being the modulus of the solid material and $p \geq 3$ the penalization factor that drives the solution towards a 0-1 design [3, 31]. The sensitivity of the objective function is expressed as:

$$\frac{\partial c}{\partial x_e} = -p(x_e)^{p-1}\mathbf{u}_e^T\mathbf{k}_0\mathbf{u}_e. \tag{4}$$

where the terms in the equation are as described above. Interest readers are referred to [30] for more details on the problem formulation. The compliance minimization problem is solved iteratively using gradient-based optimization algorithms. Sensitivity analysis (Eq. (4)) is performed to compute the derivatives of the objective function with respect to the design variables, which are used to update $\mathbf{x}$ in each iteration [28].

2.2. Deep operator networks

To create a surrogate model for the finite element function in the topology optimization process, we consider a deep operator network to approximate the finite element operator. Specifically, we propose a Convolutional Recurrent Operator Approximator Network (CRONet) to approximate the mapping from the design density $\mathbf{x}$ and applied load F inputs to the element displacement U output.

The proposed CRONet is a unique neural network structure designed to approximate nonlinear operators mapping of 2D/3D data into a 3D space. CRONet builds on the concept of deep operator networks (DeepONet), first proposed by Lu et al. in 2019 [21]. By design, operator networks are capable of learning mappings between function spaces, making them suitable for approximating operator that maps the material distribution $\mathbf{x}$ to the structural response $\mathbf{u}(x)$ [22]. Operator networks like DeepONet and CRONet are founded on the universal approximation theorem for operators, which states that neural networks can approximate any continuous operator under specific conditions, providing the theoretical foundation for these networks [8]. The universal approximation theorem is summarized as follows. Let G be an operator that maps a continuous input function u sampled at specific sensor locations $\{x_1, x_2, \ldots, x_m\}$ to the output function $G(u)$. Given any point y within the domain of $G(u)$, the output of the operator is given by $G(u)(y)$. Consequently, the network receives inputs comprised of two components, u and y, and produces an output $G(u)(y)$. The expression representing the use of deep operator networks (and by extension CRONet) for the approximation of G based on the approximation theorem for operators is given as,

$$\left| G(u)(y) - \sum_{k=1}^{p}\sum_{i=1}^{n} c_i^k\,\sigma\left(\sum_{j=1}^{m}\xi_{ij}^k u(x_j) + \theta_i^k\right)\sigma\left(w_k\cdot y + \zeta_k\right)\right| < \epsilon$$

where σ is a continuous, non-polynomial activation function, c_i^k, ξ_{ij}^k, θ_i^k, and ζ_k are constants, and $w_k \in \mathbb{R}^d$ is a weight vector.

An operator network comprises two primary subnetworks: the branch network and the trunk network. The branch network encodes the input function at specific sensor points, $\{x_1, x_2, \ldots, x_m\}$, which are locations in the input domain where the function is evaluated. This network processes discrete values of the input function $u(x_i)$ and maps them into a latent representation [22]. This encoding method effectively captures the functional input without requiring a continuous representation, which is advantageous in practical applications where only sampled data may be available.

The trunk network takes as input the coordinates y where the output operator needs to be evaluated. This network learns a mapping from spatial coordinates to the solution space. The final output of the network, $G(u)(y)$, is obtained by combining the outputs from both networks, representing the function's output at specific locations [22]. Essentially, the branch network aggregates information from the input function u sampled at sensor locations $\{x_1, x_2, \ldots, x_m\}$, while the trunk network processes the spatial coordinate y to produce an approximation of $G(u)(y)$ within an error bound ϵ. The outputs of both networks are combined through an inner product to predict the structural response at specified locations. To enhance the flexibility of the model, a bias term $b_0 \in \mathbb{R}$ can be added in the last stage to reduce the generalization error [21]. The approximation is then expressed as:

$$G(u)(y) \approx \hat{\mathbf{u}}(x) \approx \sum_{k=1}^{r} b_k(x)\,t_k(y) + b_0, \tag{5}$$

where b_k are the outputs of the branch network, $t_k(x)$ are the outputs of the trunk network, r is the number of neurons in the last layer of both subnetworks, and b_0 is an additional bias term added in the final stage.

The architecture of the operator networks enables high-accuracy modeling of complex operator mappings, especially in cases involving differential equations or dynamical systems, where traditional neural network methods may struggle with computational efficiency or scalability. It also allows the network to learn the desired operator with minimal training samples and few training iterations, as demonstrated in the case studies 3. This is an advantage of operator networks, unlike the conventional deep learning approaches that typically require significant data and training time. This architecture inherently reflects the separable structure of operator learning, where function values and locations are processed independently, thereby optimizing both storage and computation [22].

2.3. CRONet architecture for operators approximation

In the context of topology optimization, operator networks can approximate the mapping between material distribution $\mathbf{x}$ function space to the structural response $\mathbf{u}(y)$ function space, where y denotes spatial coordinates. However, the DeepONet architecture presented in [22] considers the branch network inputs as vectors and the trunk network input as either scalar or vector, depending on the dimensionality of the domain of $G(u)$. The network's output is a real number representing the operator mapping at a specific location in the output function. This

is, however, not applicable to the current topology optimization problem. The branch network input $\mathbf{x}$ is a 2D array, while the trunk network input F and the network output U are 3D arrays. This demonstrates a significant deviation from the input and output data type used in training DeepONet and thus necessitates a new architecture that can cater to the specific need (2D and 3D) of the problem domain.

To this end, we propose CRONet, a deep operator network that can approximate operators mapping high dimensional input data to another high dimensional output space. The CRONet also consists of two sub-networks, the branch and the trunk networks. As previously discussed, the input to the branch network will be the material distribution $\mathbf{x}$ over the discretized domain Ω. In contrast, the trunk network input is the load information F at each spatial coordinate where displacements are evaluated. The network output will be the displacement matrix U containing the displacement response at each spatial coordinate along all possible movement dimensions.

For a 2D design domain (the focus of this work) located on the xy-plane, the material distribution $\mathbf{x}$ -the input to the branch network- is a matrix of dimension $\mathbb{R}^{m\times n}$, where m is the number of discretizations along the x-axis and n is the number along the y-axis. The dimensions of the trunk input F and the network output U are $\mathbb{R}^{m\times n\times p}$, where p is the degree of freedom of the design domain. Given that each element in the discretization has two degrees of freedom, we have $p = 2$ for a 2D design domain. Therefore, the standard operator network requires specific configurations to meet the dimensional demands of the topology optimization use case.

The first step to creating the CRONet architecture is that we replace the standard fully connected neural network used for the branch and trunk networks with 2D and 3D convolutional neural networks (2D-CNN and 3D-CNN, respectively) (see Figure 1). This allows the CRONet network to capture spatial information within the inputs to the network. Additionally, the output of the network changes from predicting a real number, as in the vanilla architecture, to predicting a 3D displacement matrix.

Another significant modification involves including a recurrent neural network (RNN) within the branch network to capture temporal information. Since the topology optimization process iteratively improves upon the previous iterations' density distribution ($\mathbf{x}_t$), it is beneficial to leverage the temporal information available within these outputs. Therefore, to enhance the prediction of the displacement matrix, we consider using the material distribution matrices from the previous 10 steps, $\mathbf{x}_{t-10}, \cdots, \mathbf{x}_{t-1}$, to predict the displacement matrix U_t at the current time step. Combining the RNN, capturing temporal information, with the CNN capturing spatial information, we strategically design a convolutional recurrent neural network (CRNN) optimized for the current problem for the branch subnetwork in the CRONet architecture. Since the applied load remains fixed at all iterations, we only need to capture the spatial information and the applied load on each element within the design domain. Therefore, we opt for a convolutional neural network (CNN) for the trunk input. This modification allows the network to effectively learn spatial and temporal correlations, which are crucial for accurate displacement predictions in the topology optimization process. The architectural framework of the proposed CRONet operator network is presented in Fig 1

Essentially, the 2D-CNN is utilized in the branch network to process the material distribution $\mathbf{x}$, while the 3D-CNN is employed in the trunk network to incorporate the load information and evaluate displacements across all spatial coordinates. This modification enables the network to better understand the spatial correlation between neighboring elements, which is critical for accurate displacement predictions.

2.4. Training CRONet for Structural Response Prediction

To train the CRONet model for element displacement prediction, a dataset of material distributions, corresponding loads, and displacements was generated using the finite element analysis (FEA) solver. The topology optimization process for the problem at hand is first run for the first few iterations (20 were used for this work). Each training sample consists of the last 10 material distributions, $\mathbf{x_{t-10}} \cdots \mathbf{x_{t-1}}$, a load configuration F, and the corresponding displacement field U_t, that is, a data instance $D_t = (\mathbf{x_{t-10}}, \cdots, \mathbf{x_{t-1}}, F, U_t)$. The training objective of the CRONet is to minimize the difference between the predicted displacement fields and the reference FEA displacement fields. This could be achieved using any of the common loss functions L such as the mean squared error (MSE), the mean absolute percentage error (MAPE), or the Structural Similarity Index (SSIM) loss, which is specifically designed for predicting 2D or 3D arrays to emphasize structural similarity. The SSIM loss function is used in this work and is expressed as:

$$L(\theta) = 1 - \frac{1}{N}\sum_{i=1}^{N} \mathrm{SSIM}(U^{(i)}, \hat{U}^{(i)}), \tag{6}$$

where N is the number of training samples, $U^{(i)}$ is the displacement field obtained from FEA for the i-th sample, and $\hat{U}^{(i)}$ is the displacement field predicted by CRONet for the same sample. θ represents the learnable parameters of the network, which include weights and biases of the convolutional layers.

The advantage of using CRONet as a surrogate model lies in its computational efficiency. Unlike traditional FEA solvers, which can be computationally expensive due to repeated iterations required in the optimization loop, CRONet provides near-instantaneous predictions of displacements once trained. This reduces the overall computational cost of the topology optimization process, making it more feasible for large-scale problems. Furthermore, the CRONet-based topology optimization framework can be seamlessly integrated into gradient-based optimization methods. The displacement predictions from CRONet can be flattened and used to compute the sensitivity of the objective function with respect to the design variables, thereby enabling efficient gradient updates. This integration significantly accelerates the convergence of the optimization process.

The results presented in this work demonstrate that the modified DeepONet architecture can effectively approximate the FEA solver for 2D topology optimization problems. The predicted displacement fields closely match the reference FEA so-

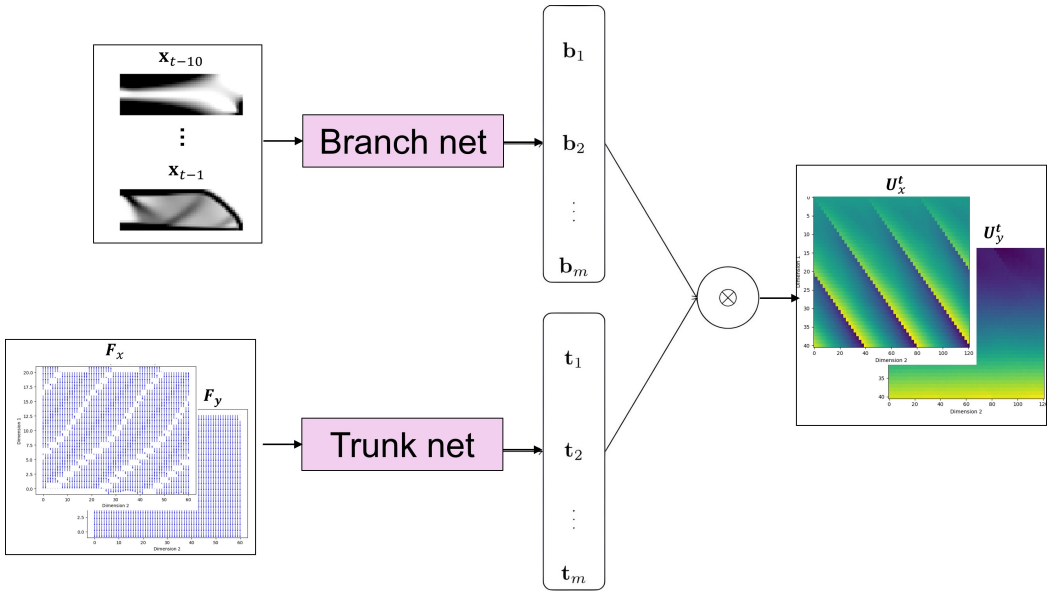


Figure 1. Proposed CRONet architecture illustrating the trunk and branch sub-networks, their inputs, and the network outputs.

lutions (see Figures 4 and 7), and the optimized designs obtained using the CRONet surrogate are comparable to those obtained using traditional FEA-based optimization (see Figure 8). This validates the potential of CRONet as a powerful tool for accelerating topology optimization in engineering applications.

The network architecture is tailored with appropriate depth and width to capture the complexity of the mapping from $\mathbf{x}$ and F to $\mathbf{u}$. Activation functions such as SiLU, tanh, and ReLU can be employed for more customized design, and techniques like batch normalization and dropout may be utilized to enhance training efficiency and prevent overfitting [18, 34]. To enhance the generalization capability of CRONet, techniques like multi-fidelity training can be employed, where both high-fidelity (fine mesh) and low-fidelity (coarse mesh) FEA data are used [23] to generate training data. Data augmentation and regularization methods can also be applied to improve performance.

2.5. Integration of CRONet into the Optimization Process

Once trained, CRONet replaces the FEA solver within the topology optimization loop. The complete integration process involves the steps which have been formalized into algorithm 1 to enhance clarity and reproducibility. By using CRONet, the computational cost per iteration is significantly reduced (see Table 1) since the expensive FEA solver is bypassed during most iterations. This allows for faster convergence and enables the handling of larger and more complex problems within reasonable computational times [27].

It is important to ensure that the predictions of CRONet remain accurate throughout the optimization process. If the material distributions encountered during optimization deviate significantly from those in the training set, the network may need retraining or online updating [15]. Adaptive sampling strategies can be employed to update the training dataset with new samples from the optimization trajectory. To validate the effectiveness of the proposed method, numerical experiments are conducted comparing the performance of the CRONet-integrated optimization with the traditional FEA-based approach in terms of computational efficiency and solution quality.

The maximum design variable change, the change in maximum variable change, and the convergence check in algorithm 1 are computed as follows:

$$\Delta_{t-1} = \max\left(|\mathbf{x}^{t-1} - \mathbf{x}^{t-2}|\right) \tag{7}$$

$$\delta = |\Delta_{t-1} - \Delta_{t-2}| \tag{8}$$

$$\Delta_t = \max\left(|\mathbf{x}^{t} - \mathbf{x}^{t-1}|\right) \tag{9}$$

Here, Equation (7) computes the maximum change in design variables between iterations $t-1$ and $t-2$. Equation (8) calculates the change in the maximum variable change between iterations, and Equation (9) is used for the convergence check.

3. Case Study

In this section, we present two topology optimization case studies to validate the performance of the proposed method (CRONet) compared to the traditional FEA approach. Specifically, we focus on two benchmark problems: (1) the Messerschmitt-Bölkow-Blohm (MBB) beam with a unit load applied at the center, and (2) the cantilever beam subjected to a downward load at the free end. These case studies are chosen because they are widely used in the literature to evaluate the effectiveness of topology optimization methods, and they provide a representative test of the capabilities of CRONet in solving both symmetric and asymmetric load scenarios. The goal is to assess how well CRONet performs in terms of computational efficiency, accuracy of structural response, and convergence characteristics when compared to the traditional finite el-

Algorithm 1 CRONet Integration for Topology Optimization

Initialization:
– Initialize material distribution $\mathbf{x}^0$ (e.g., $\mathbf{x}^0 = \mathbf{1}$).
– Set iteration counter $t = 0$, maximum iterations $T_{\max}$, and convergence threshold ϵ.
Data Generation:
for $t = 1$ to s (e.g., $s = 20$) **do**
  Compute displacement field $\mathbf{U}^t$ using the FEA solver.
  Store data $\{\mathbf{x}^t, F, \mathbf{U}^t\}$ for training.
end for
Train CRONet Model:
– Train the CRONet model using data from initial iterations.
while not converged and $t < T_{\max}$ **do**
  Increment iteration counter: $t = t + 1$.
  if $t > s$ **then**
    Compute maximum design variable change Δ_{t-1} using Eq. (7).
    Compute change in maximum variable change δ using Eq. (8).
    if $\delta > \epsilon$ **then**
      Predict Structural Response:
– Use CRONet to predict displacement field $\hat{\mathbf{U}}^t$.
    else
      Correct Response:
– Use the FEA solver to compute $\mathbf{U}^t$.
    end if
  else
    Compute Displacement Field:
– Use the FEA solver to compute $\mathbf{U}^t$.
  end if
  Sensitivity Analysis:
– Compute sensitivity of objective function w.r.t. $\mathbf{x}$ using $\mathbf{U}^t$.
  Design Update:
– Update material distribution $\mathbf{x}^{t+1}$ using an optimization algorithm (e.g., OC method).
  Convergence Check:
  Compute maximum change in design variables Δ_t using Eq. (9).
  if $\Delta_t < \epsilon$ **then**
    Terminate the optimization loop.
  end if
end while

ement analysis (FEA)-based approach and the neural-network method proposed by Chandrasekhar and Suresh [7].

3.1. System Specifications

All case studies within this work were executed on a machine running Windows 10, featuring an Intel64 Family 6 Model 165 processor with 10 physical cores and 20 threads, operating at a base frequency of 3.60 GHz. The system had a 64-bit architecture (AMD64) and was equipped with 32 GB of RAM, of which 21 GB was used, leaving 11 GB available during the runtime. Python version 3.8.19 was used for implementing and running the CRONet model. These specifications provided sufficient computational resources to effectively train and test the CRONet model and carry out the topology optimization tasks presented in this study.

3.2. Case Study I: MBB Beam Optimization

The first case study involves the optimization of the Messerschmitt-Bölkow-Blohm (MBB) beam, which is a benchmark problem widely used in topology optimization literature. The design domain for the MBB beam is rectangular, fixed at both ends, with a unit load applied at the midpoint of the top edge as shown in Fig. 2a. The MBB beam is discretized into a structured mesh with a resolution of 180×60 elements (see Fig. 2a). The boundary conditions involve fixed support at both ends, representing the symmetry of the structure, and a single point load applied at the center. Taking advantage of the symmetry of the MBB beam, the design domain can be reduced by half, as illustrated in Figure 2b. This reduction allows us to solve the optimization problem on a smaller domain, thereby improving computational efficiency while preserving the key structural characteristics. The material properties used in this optimization problem are homogeneous, with Young's modulus set to $E = 1$ and Poisson's ratio $\nu = 0.3$. The objective is to minimize the compliance (maximize stiffness) of the structure while ensuring that the optimized design maintains the desired volume fraction of 0.5.

For the CRONet model training, we initially ran the traditional topology optimization (TO) using finite element analysis (FEA) for 20 iterations to gather data, which served as the training dataset. The model was trained over 150 iterations, with the training loss plot displayed in Fig. 3. Once trained, the CRONet model was used to predict the displacement of each element, which was then compared to the FEA-generated displacements in Fig. 4, demonstrating high accuracy. Subsequently, the trained model was employed to predict responses for the remaining optimization iterations, significantly accelerating the optimization process. The optimization was performed using both the traditional FEA-based topology optimization approach and the proposed CRONet approach, with convergence criteria defined by the relative change in objective function values being below 1×10^{-2}. Parameters such as the filter radius ($r = 1.5$) and penalization factor ($p = 0.3$) were kept constant between the two methods to allow for a fair comparison.

3.3. Case Study II: Cantilever Beam Optimization

The second case study focuses on the optimization of a cantilever beam, another classic problem in structural optimization. The cantilever beam is fixed at one end and subjected to a downward load at the free end. The objective, similar to the MBB beam, is to minimize compliance subject to a volume constraint.

The design domain is discretized into 120×80 elements, with fixed boundary conditions on the left edge and a downward point load at the top-right corner. The material properties are identical to those in the MBB beam case, with Young's modulus set to $E = 1$ and Poisson's ratio $\nu = 0.3$. Similar to the MBB-beam, a CRONet model was trained using the first 20 FEA iterations data. The loss plot and the displacement predictions from

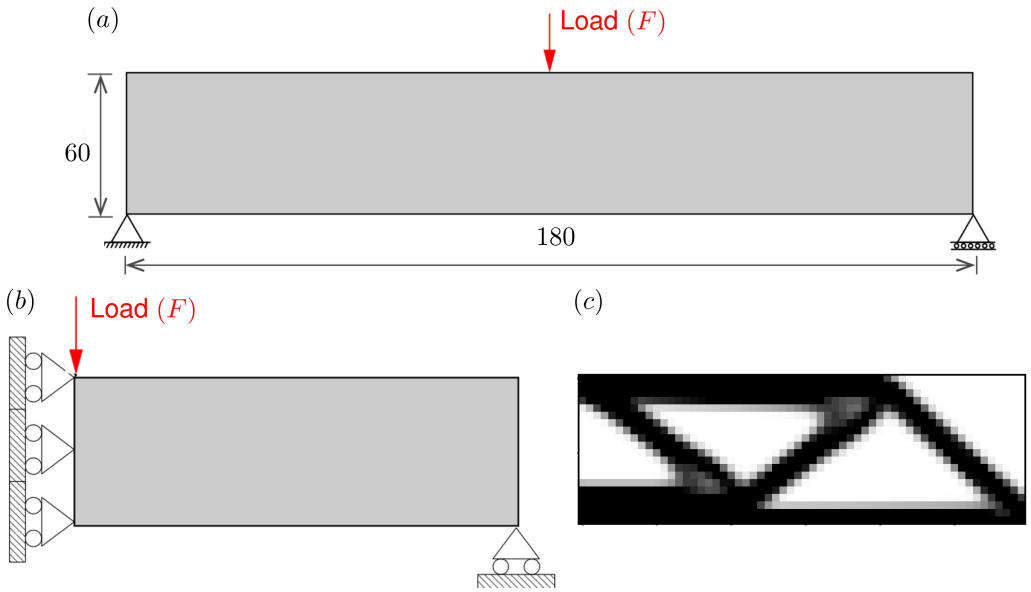


Figure 2. Topology optimization of the MBB beam. (a) The complete 2D design domain showing the beam's dimensions, applied load, and boundary conditions. (b) The half design domain with symmetry boundary conditions. (c) The topology-optimized beam for the half design domain.

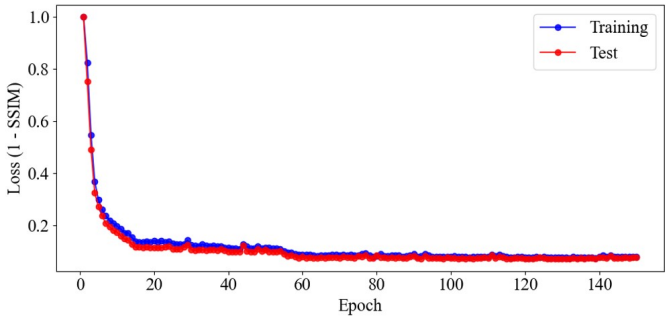


Figure 3. CRONet training and test loss over epochs for the MBB Beam. The steady decrease in both training and test loss indicates good generalization and convergence performance.

both CRONet and FEA are presented in Fig. 6 and 7. The figures illustrate the quick converges of the CRONet model within very few training iterations and predicted displacement again closely resembles the FEA output. The optimization problem was solved using both the traditional FEA-based method and CRONet, with similar convergence criteria as in the first case study.

3.3.1. Performance evaluation

To evaluate the performance of CRONet, we compared it against the traditional FEA-based topology optimization approach using several key metrics: computational time, accuracy of the structural response, and convergence rate. The computational time was measured by recording the total time taken to achieve convergence for each method. The accuracy of the structural response was evaluated by comparing the compliance values of the final designs obtained using CRONet and the traditional method. The convergence rate was assessed by analyzing the number of iterations required for each method to reach the specified convergence criteria.

The results presented in Table 1 show that CRONet significantly reduces computational time while maintaining comparable accuracy in the structural response. Specifically, CRONet converged with approximately 78% and 79% reduction in computational time for the MBB beam and cantilever beam, respectively, compared to the traditional approach, demonstrating its advantage in accelerating the optimization process. This is attributed to the surrogate model's ability to predict the displacement fields more efficiently than the traditional FEA solver, thereby reducing the computational burden of each iteration.

3.3.2. Computational Cost

Table 1 summarizes the complete computational cost of CRONet, including the setup overhead. The initial phase involves gathering 20 data instances from FEA-based iterations, which takes an average of 24 s for the first case study and 116 s for the second. Following this, model training requires roughly 150 s for the first case study and 760 s for the second. Once trained, CRONet produces predictions rapidly, resulting in a 70–80% reduction in computation time per iteration compared to traditional FEA. Overall, even after accounting for the initial overhead, the total time to solve the problem with CRONet (255 s) is notably lower than the 356 s required by the FEA-based method.

3.3.3. Accuracy and Convergence

The accuracy of the CRONet predictions was assessed by comparing the compliance of the optimized structures from CRONet and the traditional FEA-based approach. Figure 8 presents a side-by-side comparison of the final MBB Beam designs (a) and (b), along with an element-wise percentage er-

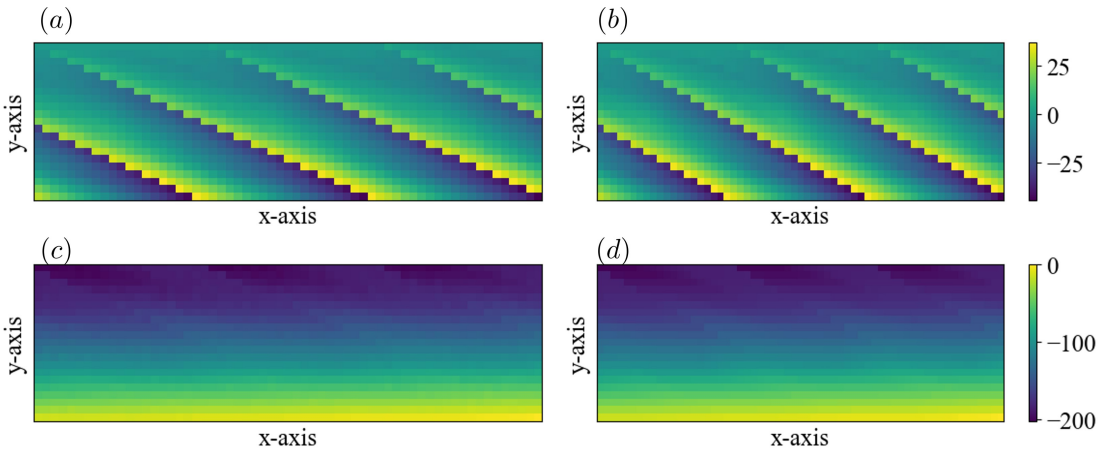


Figure 4. Comparison of predicted displacement matrices for the MBB Beam: (a) y-axis displacement from CRONet, (b) y-axis displacement from FEA, (c) x-axis displacement from CRONet, (d) x-axis displacement from FEA. The similarity between CRONet and FEA predictions indicates that CRONet maintains high accuracy in predicting structural behavior.

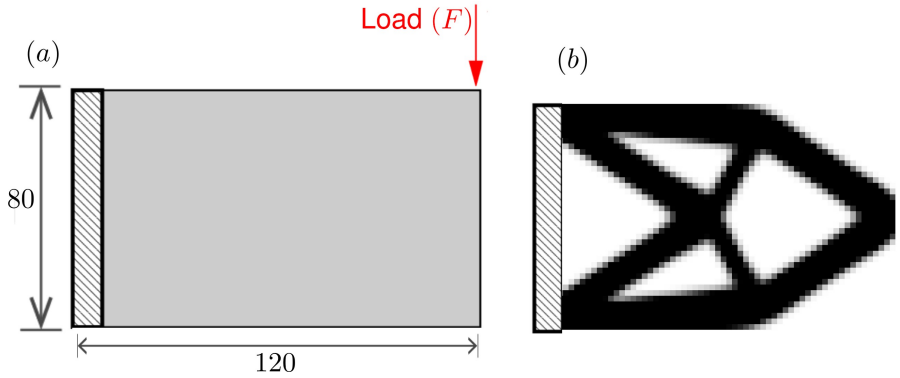


Figure 5. Topology optimization of the cantilever beam. (a) The design domain showing the beam's dimensions, applied load, and boundary conditions. (b) The topology-optimized beam.

Table 1. Comparison of optimization performance metrics between CRONet-TO, NN-TO, and FEA-TO for MBB and Cantilever beam case studies

ine Case study	Model	Total Opt. time (s)	Avg. iter. time (s)	Time reduction (%)	# of iter. until conv.	Training time (s)	Conv. err.
ine	FEA	356	1.20	–	297	–	–
MBB beam	CRONet	81	0.26	78	317	150	0.03
	NN	120	0.40	66	310	163	0.09
ine	FEA	1792	5.80	–	304	–	–
Cantilever beam	CRONet	370	0.94	79	328	760	0.05
	NN	700	1.50	61	340	846	0.12
ine							

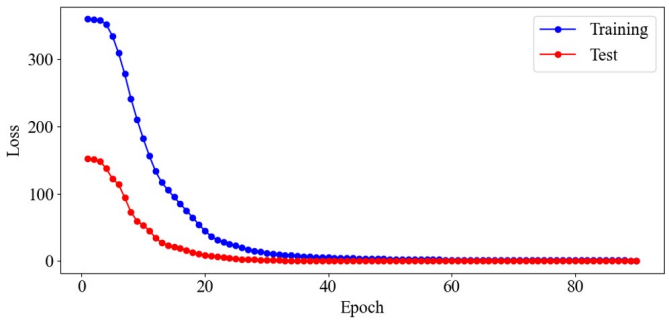


Figure 6. CRONet training and test loss over epochs for the cantilever beam. The convergence trend demonstrates CRONet's capacity for effective learning and accurate predictions.

ror map (c). CRONet successfully captures the primary load-bearing structure, with only minor deviations at boundary or high-gradient regions. These localized discrepancies do not significantly affect the overall structural performance. Moreover, the low error across most of the domain highlights CRONet's robustness and underscores its potential to greatly reduce computational costs while maintaining accuracy in large-scale or real-time topology optimization scenarios. The final designs differ by less than 5% in compliance, showing that CRONet retains high accuracy while considerably accelerating the optimization process. Additionally, CRONet converges in fewer than 25 iterations—only slightly more than the traditional method—yet the lower cost per iteration makes it well-suited

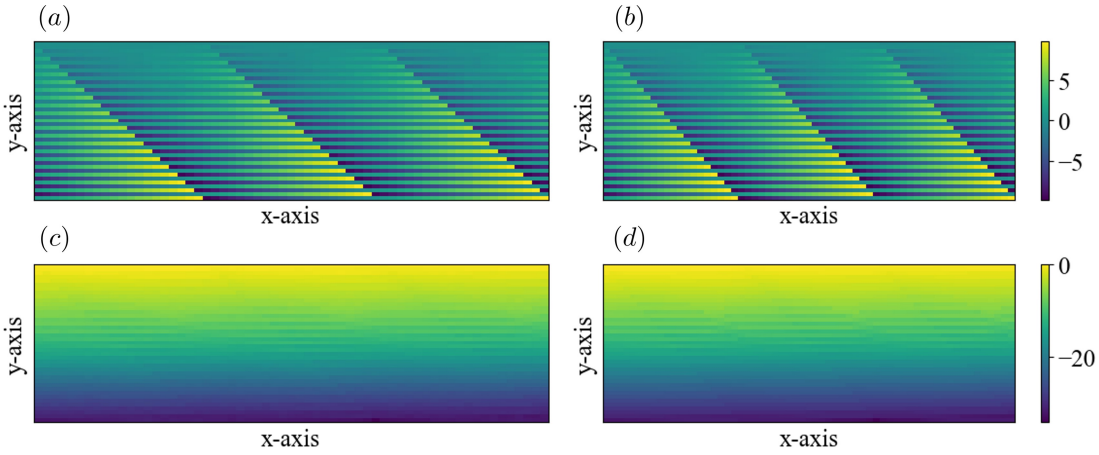


Figure 7. Comparison of predicted displacement matrices for the cantilever beam: (a) y-axis displacement from CRONet, (b) y-axis displacement from FEA, (c) x-axis displacement from CRONet, (d) x-axis displacement from FEA. The results show a high degree of similarity, indicating accurate prediction by CRONet.

for large-scale problems where computational resources are often constrained.

3.3.4. Implications for Additive Manufacturing

The results obtained from both case studies indicate that the CRONet model provides significant improvements in computational efficiency for topology optimization, which is highly beneficial for additive manufacturing (AM). The ability to optimize complex geometries with minimal computational cost supports rapid design iterations, a crucial aspect of AM workflows. As seen in Figures 4 and 7, the displacement predictions by CRONet closely match those of the traditional FEA-based approach, ensuring the reliability of the optimization process. Moreover, the reduction in computational time observed in Table 1 highlights CRONet's capability to accelerate optimization tasks without compromising the quality of the final designs. This makes CRONet particularly well-suited for real-time design updates and iterative prototyping in AM environments, where design flexibility and rapid response are essential.

Additionally, CRONet's application demonstrates its potential to handle large-scale problems effectively, supporting more ambitious design goals within additive manufacturing. The use of convolutional and recurrent layers allows the network to efficiently capture spatial and temporal dependencies, providing accurate predictions even with reduced training data, as seen in the convergence plots (Figures 3 and 6). This feature makes it a viable candidate for optimizing complex AM components, potentially reducing the gap between design and manufacturing.

4. Conclusions

This study presents a novel approach for accelerating topology optimization to enable manufacturers to adapt existing designs for individual applications using a convolutional recurrent operator approximator network (CRONet). The proposed CRONet model successfully replaces the costly finite element analysis (FEA) solver within the optimization loop, leading to a substantial reduction in computational time. The results from the MBB beam and cantilever beam case studies demonstrate that CRONet can achieve convergence at a fraction of the time required by FEA-driven optimization while maintaining the structural integrity of the optimized designs. The performance gains achieved by CRONet, including an average reduction in computational time of approximately 78% compared to traditional methods, indicate that it is well-suited for real-time design iterations in additive manufacturing. The compliance error between CRONet and FEA solutions remained within acceptable bounds, with a maximum deviation of 3%, showcasing its accuracy in structural response prediction.

Despite these promising results, some limitations exist that will be addressed in future work. The current implementation of CRONet relies on a limited dataset of FEA solutions for training. While the model shows strong generalization capabilities for the same problem specifics, further research could explore ways to enhance CRONet's performance with fewer training samples, potentially utilizing transfer learning techniques. The current method only trains CRONet to replace the FEA for the same problem; however, CRONet has the potential to generalize beyond a single problem specification, although this will require generating data from multiple problems with different loading conditions and dimensions. This is another area of future research in this domain. Finally, extending this approach to multi-physics and multi-objective topology optimization would broaden its applicability in engineering design. In conclusion, CRONet provides an effective framework for accelerating topology optimization, making it highly suitable for complex geometry optimization in additive manufacturing. By enabling rapid iterations and reducing computational costs, CRONet supports more efficient and flexible design processes, paving the way for real-time, data-driven optimization in advanced manufacturing environments.

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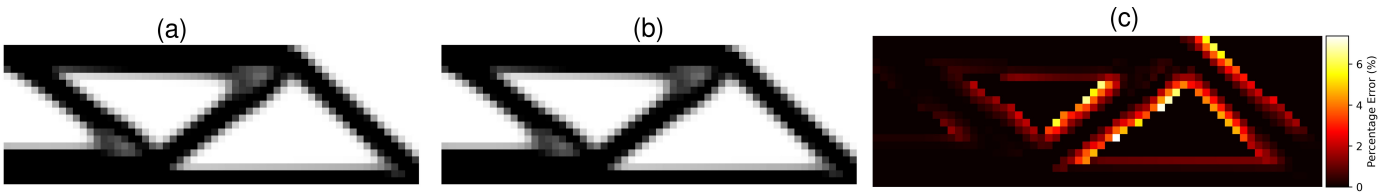


Figure 8. Comparison of final design structures for the MBB Beam obtained using CRONet and FEA. (a) Optimized design from CRONet; (b) corresponding design from traditional FEA; (c) element-by-element percentage error map between CRONet and FEA, illustrating the surrogate model's reliability.

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